

AAYUSH PRASAD

☎ +91-9328788481 ✉ dev.aayushdroid@gmail.com 🔗 linkedin.com/in/aayush-droid 📁 github.com/aayush-decoder

Education

Sardar Vallabhbhai National Institute of Technology, Surat

Bachelor of Technology in Artificial Intelligence

Aug 2024 – May 2028

CGPA: 8.63

Experience

Centre for Development of Advanced Computing (CDAC), Noida

Cyber Security Intern, CyberGyan (MeitY, Govt. of India)

Jun 2025 – Jul 2025

Certificate

- Built security assessment workflows using **Nmap**, **Nessus**, **Wireshark**, **Amass**, and **Kali Linux** to analyze simulated enterprise networks.
- Identified **vulnerabilities**, **exposed services**, and **security misconfigurations** through enterprise network reconnaissance.
- Engineered heuristic-based **detection workflows** by analyzing **packet captures**, **system logs**, and **process behavior** to detect malicious activity.

Projects

TransitOps | *Live* | *GitHub*

Jul 2026

Technologies: *Next.js, TypeScript, PostgreSQL, Prisma ORM, Auth.js, Redis, Tailwind CSS*

- Architected a **role aware transport operations platform** spanning **8 operational modules**, **4 user roles**, and **35+ REST APIs**, implementing RBAC, transactional workflows, and lifecycle automation for fleet, trip, maintenance, finance, and safety operations.
- Optimized the **PostgreSQL and Prisma backend** by introducing composite indexes, selective query projections, and Redis caching, reducing API latency by **72%** and improving operational dashboard response time from **620 ms to 95 ms (6.5x)**.
- Engineered scalable dashboard pipelines with asynchronous data fetching, cursor-based pagination, and OSRM route caching, improving page load time by **48%**, supporting **100k+ fleet records**, and reducing routing requests by **82%**.

HFT Market Analysis & Trading Platform | *GitHub*

Dec 2025 – Jan 2026

Technologies: *C++, FastAPI, gRPC, PostgreSQL, PyTorch, React, WebSockets, Docker, AWS*

- Engineered a **low-latency distributed analytics platform** to process limit order book data, enabling real-time detection of **spoofing and liquidity gaps** in high-frequency markets.
- Built a high-performance **C++ analytics engine** for real-time signal processing, reducing end-to-end latency by **85%**.
- Designed a **scalable ingestion pipeline** capable of processing **1.2B+ data points** while sustaining **50,000+ L2 updates/sec** for high-volatility backtesting workloads.
- Integrated a **DeepLOB-based PyTorch inference service** into the analytics pipeline, improving short-term price prediction accuracy by **14%** using 100-level order book data.

Webify | *GitHub*

Jan 2026 – Feb 2026

Technologies: *TypeScript, Node.js, AWS (S3, SQS, DynamoDB), Express.js, RabbitMQ*

- Reduced deployment time from **4 min to 45 sec** by implementing **asynchronous build execution**.
- Achieved under **150ms API latency** by eliminating request blocking using **AWS SQS-based job queues**
- Handled **concurrent deployments** and reduced backpressure by decoupling compute-intensive build tasks
- Engineered subdomain-based deployments by implementing **stream-based asset delivery** from S3 with host-based routing

Technical Skills

Languages: Java, C, C++, Python, JavaScript, TypeScript, SQL, HTML, CSS

Computer Science Fundamentals: Data Structures & Algorithms, Object Oriented Programming, Database Management System (DBMS), Operating Systems, Computer Networks, Problem Solving, Low Level System Design

Frameworks & Development: React, Next.js, Node.js, FastAPI, Django, REST APIs

Databases & Tools: PostgreSQL, Redis, RabbitMQ, Git, GitHub, Docker, CI/CD Pipelines

Cloud & Infrastructure: AWS, Machine Learning, Data Science, PyTorch, Scikit-Learn

Achievements

1st Rank Winner, Echelon 2026, a National Level hackathon. Built a Multimodal Misinformation Detection pipeline

Winner, Adani Powermind, secured 1st rank in the hackathon in 'Energy Forecasting using AI' track.

Top 5, TRACS at WASP 2025, ranked 4th on the official Kaggle leaderboard for astrophysical document classification

Positions of Responsibility

Developer, Association for Computing Machinery (ACM), NIT Surat

Mentor, Nexus, NIT Surat - Guided juniors in problem-solving and DSA.

May 2026 – Present

Jan 2026 – Present